

HAKIM AREZKI, PhD

DATA & BI CONSULTANT — POWER BI, SNOWFLAKE, DATAIKU, ETL

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Data and BI consultant with 12 years of experience (15 including doctoral research) designing, migrating and operating enterprise Data and BI platforms in regulated banking and financial environments (BIL, BPCE) and large industrial groups (TotalEnergies, RATP).

Modernisation of legacy decision-support estates towards the cloud: inventory and rationalisation of the reporting portfolio, redesign of data models, industrialisation of transformation pipelines, semantic layer rebuild, Row-Level Security, reconciliation and user enablement.

Command of the full chain from ingestion to reporting: ETL and ELT, modelling, trusted data layers, semantic models, dashboards and business KPIs, self-service and BI governance.

Data analysis and value creation: definition and tracking of business indicators, exploratory analysis, predictive models and automation of analytical processing.

Core skills

Power BI and reporting

Semantic models, Power Query and DAX measures · Business KPIs and operational dashboards · Row-Level Security and governed access · Power BI Cloud administration: workspaces, capacities, deployment pipelines, gateways, refresh strategy · Migration of legacy BI estates from IBM Cognos · Self-service enablement, BI governance and dataset certification

ETL, ELT and data engineering

Batch and streaming ingestion frameworks · Industrialisation of transformation pipelines with dbt and Spark · Orchestration with Airflow · Real-time ingestion with Kafka · ETL tooling: Talend, Sqoop · Data modelling and trusted curated layers · Performance and cost optimisation

Snowflake and data platforms

Enterprise data platform architecture on Snowflake · Modelling and trusted data layers · Warehouse sizing, resource monitors, credit consumption tracking, query optimisation and cost allocation per business domain · Databricks, PySpark and distributed processing

Dataiku and analytics

Design and industrialisation of analytical flows in Dataiku · Moving prototypes to production · Exploratory and statistical analysis · Predictive models: classification, forecasting, anomaly detection, churn and recommendation · Automation of recurring analytical processing

Data quality, governance and compliance

Data quality, reconciliation and controls · Master Data Management · Catalog, data dictionary and lineage with Apache Atlas · RBAC access control and Row-Level Security, masking and encryption · GDPR, DORA, auditability and traceability

Business engagement and delivery

Requirements gathering and analysis · Translation into reusable data products · User training and enablement · Agile delivery: Scrum, Kanban, SAFe, Jira, Confluence

Technology environment

BI: Power BI Cloud, semantic models, Power Query, DAX, Row-Level Security, workspaces, deployment pipelines · IBM Cognos as source estate · Grafana · Data platforms: Snowflake, Databricks, dbt, Azure, GCP, AWS · Analytics: Dataiku, Python, scikit-learn, H2O · ETL and integration: Talend, Sqoop, Airflow, Kafka, Redis Streams, REST APIs · Languages and processing: Python, SQL, Spark, PySpark, R · Databases: Snowflake, PostgreSQL, Oracle SQL, MongoDB, Cassandra, HBase, Redis · Search and logs: Elasticsearch, Logstash, Kibana · Governance: MDM, data quality, lineage, catalog, Apache Atlas, RBAC, RLS, GDPR · DevOps: Docker, Kubernetes, GitLab CI/CD, Azure DevOps

Professional experience

Data & BI Lead — BIL, Banque Internationale à Luxembourg

January 2026 to present

Modernisation of the bank's decision-support estate and operation of the enterprise Data and AI platform on a Snowflake foundation.

Led the migration of the legacy IBM Cognos estate to Snowflake and Power BI: inventory and rationalisation of the portfolio of reports, packages, data models and business rules.

Redesigned data models and transformation pipelines feeding the BI estate.

Rebuilt semantic models in Power BI: Power Query, DAX measures, Row-Level Security.

Ran data reconciliation and performance testing between the legacy and target estates.

Supported and trained users, and established BI governance.

Designed and evolved Snowflake capabilities providing governed, trusted data to analytics, Machine Learning and AI use cases.

Industrialised Data Science and AI use cases using Dataiku and Snowflake, from prototype to production.

Ensured compliance: RBAC, Row-Level Security, GDPR, encryption, traceability and auditability, in line with banking requirements including DORA.

Worked with Data Engineering teams on ingestion, transformation and integration pipelines feeding both analytics and AI workloads.

Lead Data & Analytics Platform — TotalEnergies

December 2020 to December 2025

Enterprise Data Platform on Snowflake serving multiple business domains: retail, human resources and industrial operations.

Designed and maintained Power BI datasets, dashboards and KPIs used by business teams for operational monitoring and decision-making.

Enabled self-service analytics on governed datasets.

Designed and industrialised scalable data pipelines with Databricks, PySpark, dbt, Airflow and Kafka, covering batch and streaming ingestion.

Worked on Snowflake data architecture, data modelling and trusted data layers supporting analytics, reporting, operational processes and AI use cases.

Translated retail and HR business requirements into reusable data products.

Contributed to Master Data Management initiatives and implemented data quality, governance, lineage and access-control practices.

Industrialised Machine Learning use cases on Databricks and MLflow, from experimentation to production.

Coordinated multi-disciplinary teams of around 20 people and managed external service providers.

Head of Data — RATP, MSM programme with Alstom, Renault and IRT SystemX

August 2018 to December 2020

Built and deployed a multi-layered Data Lake on Azure for scalable ingestion, processing and analytics.

Developed and maintained Power BI dashboards, custom web UIs and operational KPIs.

Developed predictive models for passenger flow, incident duration and operational planning.

Built OCR and computer-vision pipelines for automated processing of unstructured data.

Implemented anomaly detection and log analysis to ensure data quality and regulatory compliance.

Trained business and technical teams on data visualisation and process automation.

Lead Data Science & Governance — BPCE, Banking & Insurance

January 2017 to August 2018

Designed and deployed the enterprise data dictionary and lineage capability on Apache Atlas, establishing traceability of financial data assets through to regulatory reporting.

Structured metadata, ownership and data definitions across compliance, sanctions and fraud domains.

Defined detection-system performance indicators and delivered Power BI dashboards for risk and compliance teams.

Significantly reduced false positives through confusion-matrix analysis and classification models.

Deployed Elasticsearch and Kibana clusters for large-scale log indexing and alert investigation.

Ensured GDPR compliance of financial data processing chains.

Lead Data Scientist — Luxury Retail

September 2015 to January 2017

Integrated CRM and transactional data from Oracle SQL to Cassandra via Sqoop, captured real-time streams with Kafka, and enriched internal data with external datasets.

Defined and tracked retail KPIs — sales by category, return rates, churn, store visit frequency — with real-time reporting through Spark, Storm and Power BI.

Built a customer intelligence engine: classification, behaviour prediction, churn and recommendation.

Data Scientist — Beamlabs / Française des Jeux

November 2014 to September 2015

Built a Big Data foundation for real-time marketing: distributed Hadoop and HBase architecture, Kafka streaming ingestion, Talend ingestion into Elasticsearch.

Developed a real-time recommendation engine and predictive models from behavioural analysis.

Enriched internal data through distributed web crawling: weather, geolocation, socio-demographic data.

R&D Engineer, PhD — CentraleSupélec and École Polytechnique

February 2010 to October 2014

Applied Machine Learning to photovoltaic cell optimisation and transistor modelling.

Developed simulation and data analysis algorithms in Python and R, with scientific visualisation in R-Shiny.

Education

PhD, CentraleSupélec, Université Paris-Saclay, Orsay (2011-2014)

Master of Sciences, Université Pierre et Marie Curie, Paris VI (2010-2011)

Master's degree in Engineering Sciences, Université Paris Diderot, Paris VII (2009-2010)

Languages

French: native

English: fluent professional proficiency

German: notions

Publications and speaking

Architecture of a Public Transport Supervision System Using Hybridization Models Based on Real and Predictive Data, IEEE / DSD 2020, pp. 440-446

Disruption Impact on the Offer and Demand on the Public Transport Network in Paris

Big Data Paris 2018, speaker: "We can tell you by what external context your sales are influenced"

Training delivered

Advanced Python for Machine Learning (SQLI Institute, 2019) · Datalab (Saagie, 2019) · SQL for Data Science (Fitec, 2018) · ELK (Fitec, 2017) · Kafka (Air4Data / SQLI Institute, 2017) · DataViz, Tableau and Power BI (SQLI Institute, 2017)